

Demonstration of Proposed Features

Date	30 June 2026
Team ID	
Project Name	Rising water
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	User Registration	Allows new users to create accounts with hashed passwords stored in PostgreSQL.	Implemented	Yes	Werkzeug password hashing ensures security.
2	Secure User Login	Authenticates users via email/password and manages Flask sessions.	Implemented	Yes	Session-based authentication with redirect support.
3	Flood Prediction (ML Inference)	Accepts meteorological inputs and returns flood probability via XGBoost model.	Implemented	Yes	Core feature; processes inputs in milliseconds.
4	Prediction History Dashboard	Displays all past predictions for the logged-in user from PostgreSQL.	Implemented	Yes	Flask-Caching applied for faster load times.
5	Health Check API	A /health endpoint for monitoring backend uptime and routing status.	Implemented	Yes	Returns JSON status for system monitoring.

6	Reports Page	Displays ML model evaluation metrics and classification data.	Implemented	Yes	Accessible via the /reports route.
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Feature Implementation Summary:

Metric	Value
Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate (%)	100%